

1. Define data Mining. List two basic data mining tasks. Explain the KDD process and the role of data preprocessing

Data Mining:

Data mining is the process of extracting useful patterns, trends, and knowledge from large databases using statistical and machine learning techniques.

Two basic data mining tasks:

- * classification
- * clustering

KDD process:

1. Data cleaning - Removes errors and missing values.
2. Data integration - combines data from different sources.
3. Data selection - selects relevant data
4. Data Transformation - converts data into a suitable format.
5. Data Mining - Extracts useful patterns.

Role of Data preprocessing:

- * Improves data quality.
- * Removes noise and duplicates.
- * Handles missing values.
- * Increases the accuracy and efficiency of data mining.

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b. Data cleaning

Definition:

Data cleaning is the process of correcting incomplete, inconsistent, duplicate, or incorrect data.

Two data quality issues:

- * Missing values

- * Noisy data

Handling Missing values:

- * Replace with mean, median or mode.

- * Predict using machine learning.

Handling Noisy Data:

- * Binning

- * Regression

- * clustering

2.

a. Data Integration

Definition:

Data integration is the process of combining data from multiple sources into a single consistent dataset.

challenges:

- * Data redundancy

- * Data inconsistency

Handling Redundancy:

- * Remove duplicate records.

- * use correlation analysis

Handling inconsistency:

- * standardize formats.

- * Resolve conflicting values using data cleaning techniques.

b. covariance Analysis

covariance analysis measures the relationship between two variables.

Population Formula:

$$\text{cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

sample Formula

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Interpretation:

- * Positive covariance $\rightarrow$ variables increase together
- * Negative covariance $\rightarrow$ one increases while the other decreases.

3. a. stock Analysis

Given data for stocks A and B, the covariance is positive.

Answer:

Since the covariance is positive, both stock prices move in the same direction. Therefore, the stocks rise or fall together and are not independent.

b. Architecture of Data Mining system

components:

- * Data sources
- * Database / Data warehouse
- * Database server
- * Data Mining Engine
- * Pattern Evaluation Module
- * Knowledge Base
- * User Interface

Working: Data is collected from different sources, stored in the warehouse, processed by the mining engine, evaluated for useful patterns, and displayed to the user through the interface.

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a) covariance calculation

Given

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{sample covariance} = -0.15 \text{ (approximately)}$$

Result:

The covariance is negative, indicating a weak inverse relationship between X and Y .

b. Data pre-processing

Major steps:

1. Data cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization.

Pipeline:

- * Remove missing values
- * Detect and remove outliers
- * Standardize data formats
- * Normalize the data
- * Reduce unnecessary features.

This sequence improves data quality and mining performance.

5. a. Equal-Frequency (Equal-depth) Binning

The given age data is divided into 3 equal-frequency bins.

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

For smoothing, each value in a bin can be replaced by the bin mean.

b. Data Discretization

Definition:

Data discretization converts continuous numerical values into intervals (bins).

Importance:

- * Reduces noise.
- * Improves data analysis.
- * speeds up data mining.
- * Simplifies classification.

Comparison:

- * Equal-width Binning: Equal interval size; simple but sensitive to outliers.
- * Equal-Frequency Binning: Equal number of records in each bin; better for unevenly distributed data.